

Week 3

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pranav_chowdary@Pranav: ~$ nano q3.c
pranav_chowdary@Pranav: ~$ gcc q3.c -o q3
pranav_chowdary@Pranav: ~$ ./q3
Parent PID: 1110
Child PID: 1111
Child PID: 1111
Parent PID: 1110
ps -ef | grep q3
Child finished
Parent finished
pranav_chowdary@Pranav: ~$ ps -ef | grep q3
pranav_+ 1128 401 0 08:43 pts/0 00:00:00 grep --color=auto q3
pranav_chowdary@Pranav: ~$ ps -l
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
4 S 1000 401 400 0 80 0 - 1567 do_wai pts/0 00:00:00 bash
0 R 1000 1142 401 0 80 0 - 1798 - pts/0 00:00:00 ps
pranav_chowdary@Pranav: ~$ top
top - 08:43:52 up 5 min, 1 user, load average: 0.08, 0.05, 0.01
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7792.7 total, 6570.2 free, 591.1 used, 787.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7201.6 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 1154 pranav_+  20   0   8160  5984  3888 R   0.3   0.1   0:00.01 top
    1 root      20   0  24556 15524 11612 S   0.0   0.2   0:01.39 /sbin/init
    2 root      20   0   3180  2208  2072 S   0.0   0.0   0:00.10 /init
    6 root      20   0   3180  2044  1940 S   0.0   0.0   0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8+
   49 root      19  -1  50392 16428 15156 S   0.0   0.2   0:00.27 /usr/lib/systemd/systemd-journald
   89 systemd+  20   0  22540 14604 11988 S   0.0   0.2   0:00.06 /usr/lib/systemd/systemd-resolved
   93 root      20   0  35460 12456  9260 S   0.0   0.2   0:00.71 /usr/lib/systemd/systemd-udev
  172 root      20   0   2888  1952  1820 S   0.0   0.0   0:00.05 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh +
  173 root      20   0   4472  3148  2900 S   0.0   0.0   0:00.00 /usr/sbin/cron -f -P
  174 message+  20   0   8820  5420  4728 S   0.0   0.1   0:00.05 @dbus-daemon --system --address=systemd: --nofork --+
  181 root      20   0  44092 29724 14584 S   0.0   0.4   0:00.20 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-
```

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top - 08:43:52 up 5 min, 1 user, load average: 0.08, 0.05, 0.01
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
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MiB Mem : 7792.7 total, 6570.2 free, 591.1 used, 787.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7201.6 avail Mem

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 1154 pranav_+  20   0   8160  5984  3888 R   0.3   0.1   0:00.01 top
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   89 systemd+  20   0  22540 14604 11988 S   0.0   0.2   0:00.06 /usr/lib/systemd/systemd-resolved
   93 root      20   0  35460 12456  9260 S   0.0   0.2   0:00.71 /usr/lib/systemd/systemd-udev
  172 root      20   0   2888  1952  1820 S   0.0   0.0   0:00.05 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh +
  173 root      20   0   4472  3148  2900 S   0.0   0.0   0:00.00 /usr/sbin/cron -f -P
  174 message+  20   0   8820  5420  4728 S   0.0   0.1   0:00.05 @dbus-daemon --system --address=systemd: --nofork --+
  181 root      20   0  44092 29724 14584 S   0.0   0.4   0:00.20 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-+
  252 root      20   0  18428  9404  8168 S   0.0   0.1   0:00.09 /usr/lib/systemd/systemd-logind
  254 root      20   0 1857652 15512 12552 S   0.0   0.2   0:00.12 /usr/libexec/wsl-pro-service
  263 syslog    20   0 220548  5920  4644 S   0.0   0.1   0:00.07 /usr/sbin/rsyslogd -n -iNONE
  294 _chrony    20   0  24248 11808  7228 S   0.0   0.1   0:00.05 /usr/sbin/chronyd -n -F 1 -x
  319 _chrony    20   0  12092  2420  1776 S   0.0   0.0   0:00.00 /usr/sbin/chronyd -n -F 1 -x
  353 root      20   0 123456 32628 18024 S   0.0   0.4   0:00.17 /usr/bin/python3 /usr/share/unattended-upgrades/unat+
  356 root      20   0   5492  2712  2524 S   0.0   0.0   0:00.01 /usr/sbin/agetty --noreset --noclear --issue-file=/e+
  399 root      20   0   3188  1112  980 S   0.0   0.0   0:00.00 /init
  400 root      20   0   3204  1252  1108 S   0.0   0.0   0:00.05 /init
  401 pranav_+  20   0   6268  5532  3864 S   0.0   0.1   0:00.11 -bash
  402 root      20   0   8644  5516  4676 S   0.0   0.1   0:00.01 login -- pranav_chowdary
  445 pranav_+  20   0  22752 13344 10868 S   0.0   0.2   0:00.12 /usr/lib/systemd/systemd --user
  449 pranav_+  20   0  22912  4092  2088 S   0.0   0.1   0:00.00 (sd-pam)
  477 pranav_+  20   0   6136  5416  3864 S   0.0   0.1   0:00.04 -bash
  942 root      20   0 340140 22384 19328 S   0.0   0.3   0:00.03 /usr/libexec/packagekitd
  949 polkitd    20   0 306444  8492  7676 S   0.0   0.1   0:00.06 /usr/lib/polkit-1/polkitd --no-debug --log-level=not+
 1159 root      20   0   3546  6576  3348 S   0.0   0.1   0:00.00 (udev-worker)
```

```
pranav_chowdary@Pranav: ~  
GNU nano 8.7.1 q3.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int pid = fork();  
  
    if (pid == 0)  
    {  
        printf("Child PID: %d\n", getpid());  
        printf("Parent PID: %d\n", getppid());  
  
        sleep(10);  
  
        printf("Child finished\n");  
    }  
    else  
    {  
        printf("Parent PID: %d\n", getpid());  
        printf("Child PID: %d\n", pid);  
  
        sleep(5);  
        wait(NULL);  
  
        printf("Parent finished\n");  
    }  
  
    return 0;  
}  
  
[ Read 30 lines ]  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^I Execute    ^C Location   M-U Undo      M-A Set Mark  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-G Copy
```